

Variational Inference Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning
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PART A: CONCEPTUAL QUESTIONS (Simple to Intermediate)

A1. What does Shannon entropy measure, and why is the uniform distribution the maximum entropy distribution over a finite support?

Answer:

Shannon entropy measures the **average uncertainty** (or expected surprise) in the outcomes of a random variable. For a discrete random variable X taking values $x_1, \dots, x_n$ with probabilities p_i :

$$H(X) = - \sum_{i=1}^n p_i \log p_i.$$

- If X is deterministic (one outcome has probability 1), then $H(X) = 0$.
- High entropy means outcomes are hard to predict; low entropy means outcomes are more predictable.

The uniform distribution $p_i = 1/n$ maximizes entropy because any deviation from uniformity makes some outcomes more predictable and reduces entropy. Formally, one can show this using Lagrange multipliers: maximizing $H(X) = - \sum p_i \log p_i$ subject to $\sum p_i = 1$ and $p_i \geq 0$ yields $p_i = 1/n$ for all i , giving $H(X) = \log n$.

A2. Why is the KL divergence not a true distance metric? State at least two properties it lacks.

Answer:

KL divergence is not a true metric because:

1. **Asymmetry:** In general $\text{KL}(P\|Q) \neq \text{KL}(Q\|P)$. A true metric requires $d(P, Q) = d(Q, P)$.
2. **Triangle inequality not satisfied:** A metric requires $d(P, R) \leq d(P, Q) + d(Q, R)$, which KL divergence does not guarantee.

Additionally, KL divergence can be infinite (when $q(x) = 0$ for some x where $p(x) > 0$), which is unusual for a distance. However, it does satisfy non-negativity ($\text{KL}(P\|Q) \geq 0$) and identity ($\text{KL}(P\|Q) = 0 \iff P = Q$).

A3. Explain the difference between forward KL $\text{KL}(p\|q)$ and reverse KL $\text{KL}(q\|p)$.

Answer:

- **Forward KL** $\text{KL}(p\|q) = \mathbb{E}_p[\log p - \log q]$: The expectation is under p . This penalizes q for assigning low probability to regions where p has significant mass. The result is **mass-covering** behavior— q will try to spread out to cover all modes of p .
- **Reverse KL** $\text{KL}(q\|p) = \mathbb{E}_q[\log q - \log p]$: The expectation is under q . This penalizes q for placing mass where p is small. The result is **mode-seeking** behavior— q will concentrate on a single mode of p .

When p is bimodal (e.g., $p(x) = 0.5\mathcal{N}(-2, 0.5^2) + 0.5\mathcal{N}(2, 0.5^2)$) and q is a single Gaussian:

- Minimizing forward KL gives a broad q centered between the modes, covering both but fitting neither well.
- Minimizing reverse KL gives q that locks onto one of the two modes, completely ignoring the other (**mode dropping**).

A4. Why does variational inference minimize the reverse KL?

Answer:

VI chooses $q_\lambda(z)$ to minimize $\text{KL}(q_\lambda(z)\|p(z|x))$. This direction is preferred because:

1. The expectation is under q_λ , from which we *can* sample.
2. We can evaluate $\log p(x, z)$ (the joint model) up to a constant.
3. This leads to the ELBO objective, which is tractable to optimize.

The forward KL $\text{KL}(p(z|x)\|q)$ would require expectations under the true posterior $p(z|x)$, which is precisely the intractable distribution we are trying to approximate. We cannot sample from it or evaluate it.

The consequence of using reverse KL is that VI can exhibit mode-seeking behavior and may drop modes of a multimodal posterior.

A5. What is the mean-field approximation?

Answer:

The mean-field approximation assumes the variational distribution factorizes as a product of independent factors over the latent variables:

$$q(z) = \prod_{i=1}^m q_i(z_i).$$

Key assumption: All latent variables are independent under q .

Why it is useful:

- High-dimensional integrals factor into products of 1D integrals.
- In conjugate models, each factor q_i^* has a closed-form update.
- Enables efficient coordinate ascent variational inference (CAVI).

Main limitation: Mean-field cannot capture posterior correlations. If the true posterior has strong dependencies between variables (e.g., tilted elliptical contours), the mean-field approximation (axis-aligned ellipses) will underestimate uncertainty and miss the joint structure. This leads to biased uncertainty estimates.

PART B: MATHEMATICAL DERIVATIONS

B1. Derive the Evidence Lower Bound (ELBO) using Jensen's inequality.

Answer:

Start from the log-evidence:

$$\log p(x) = \log \int p(x, z) dz.$$

Introduce any distribution $q(z)$ by multiplying and dividing:

$$\log p(x) = \log \int q(z) \frac{p(x, z)}{q(z)} dz = \log \mathbb{E}_{q(z)} \left[\frac{p(x, z)}{q(z)} \right].$$

Since log is concave, apply Jensen's inequality $\log(\mathbb{E}[Y]) \geq \mathbb{E}[\log Y]$:

$$\log p(x) \geq \mathbb{E}_{q(z)} \left[\log \frac{p(x, z)}{q(z)} \right] = \mathbb{E}_{q(z)} [\log p(x, z)] - \mathbb{E}_{q(z)} [\log q(z)].$$

Define the ELBO:

$$\boxed{\mathcal{L}(q) := \mathbb{E}_{q(z)} [\log p(x, z)] - \mathbb{E}_{q(z)} [\log q(z)]}$$

so that $\log p(x) \geq \mathcal{L}(q)$.

Note that $-\mathbb{E}_{q(z)} [\log q(z)] = H(q)$ is the entropy of q , so equivalently $\mathcal{L}(q) = \mathbb{E}_{q(z)} [\log p(x, z)] + H(q)$.

B2. Show that maximizing the ELBO is equivalent to minimizing $\text{KL}(q(z) \| p(z | x))$.

Answer:

Start from the KL divergence:

$$\begin{aligned} \text{KL}(q(z) \| p(z | x)) &= \mathbb{E}_q \left[\log \frac{q(z)}{p(z | x)} \right] \\ &= \mathbb{E}_q [\log q(z)] - \mathbb{E}_q [\log p(z | x)]. \end{aligned}$$

Apply Bayes' rule: $p(z | x) = p(x, z)/p(x)$, so $\log p(z | x) = \log p(x, z) - \log p(x)$:

$$\text{KL}(q \| p(\cdot | x)) = \mathbb{E}_q [\log q(z)] - \mathbb{E}_q [\log p(x, z)] + \log p(x).$$

Rearrange:

$$\log p(x) = \underbrace{\mathbb{E}_q [\log p(x, z)] - \mathbb{E}_q [\log q(z)]}_{\mathcal{L}(q)} + \text{KL}(q(z) \| p(z | x)).$$

$$\boxed{\log p(x) = \mathcal{L}(q) + \text{KL}(q(z) \| p(z | x))}$$

Since $\log p(x)$ is a constant (independent of q) and $\text{KL} \geq 0$:

- Maximizing $\mathcal{L}(q)$ is equivalent to minimizing $\text{KL}(q(z) \| p(z | x))$.
- The ELBO is tight (equals $\log p(x)$) if and only if $q(z) = p(z | x)$.

B3. Derive the alternative form of the ELBO.

Answer:

Decompose the joint: $\log p(x, z) = \log p(x | z) + \log p(z)$. Substitute into the ELBO:

$$\begin{aligned}\mathcal{L}(q) &= \mathbb{E}_{q(z)}[\log p(x, z)] - \mathbb{E}_{q(z)}[\log q(z)] \\ &= \mathbb{E}_{q(z)}[\log p(x | z) + \log p(z)] - \mathbb{E}_{q(z)}[\log q(z)] \\ &= \mathbb{E}_{q(z)}[\log p(x | z)] + \mathbb{E}_{q(z)}[\log p(z) - \log q(z)] \\ &= \mathbb{E}_{q(z)}[\log p(x | z)] - \text{KL}(q(z) \| p(z)).\end{aligned}$$

$$\mathcal{L}(q) = \mathbb{E}_{q(z)}[\log p(x | z)] - \text{KL}(q(z) \| p(z))$$

Interpretation:

- $\mathbb{E}_{q(z)}[\log p(x | z)]$: **Expected log-likelihood** under the variational posterior. This term encourages q to place mass on latent values z that explain the observed data x well.
- $\text{KL}(q(z) \| p(z))$: **Complexity penalty**. This term penalizes q for deviating from the prior. It acts as a regularizer.
- VI trades off **data fit** (first term) vs. **regularization toward the prior** (second term).

PART C: PARAMETRIC ANALYSIS

C1. What happens to the optimal $q^*(z)$ as N increases?

Answer:

In the alternative ELBO $\mathcal{L}(q) = \mathbb{E}_q[\log p(x | z)] - \text{KL}(q(z) \| p(z))$:

- The expected log-likelihood $\mathbb{E}_q[\log p(x | z)] = \sum_{n=1}^N \mathbb{E}_q[\log p(x_n | z)]$ grows linearly with N .
- The KL penalty $\text{KL}(q(z) \| p(z))$ remains a fixed-size term (does not scale with N).

As N increases:

- The data term dominates, so $q^*(z)$ is primarily driven by the likelihood rather than the prior.
- The optimal q^* concentrates around the maximum likelihood estimate (its variance shrinks as $\approx 1/N$).
- The prior becomes less influential—a Bayesian “washing out” of the prior.
- Predictions become more confident.

When N is small, the prior term has a strong regularizing effect, pulling q toward the prior distribution $p(z)$.

C2. How does the quality of the mean-field approximation change with posterior correlation?

Answer:

The mean-field assumption $q(z) = \prod_i q_i(z_i)$ forces q to have a diagonal (axis-aligned) covariance. If the true posterior has strong correlations:

- The true posterior has tilted elliptical contours (off-diagonal covariance entries).
- The mean-field q can only produce axis-aligned ellipses.
- As correlations grow stronger, the mismatch between q and $p(z | x)$ increases, and $\text{KL}(q||p)$ becomes larger.
- The mean-field approximation systematically **underestimates posterior uncertainty**: marginal variances may be captured, but the joint structure is lost.
- Credible regions based on q will be too small and poorly oriented.

Alternatives that handle correlations include full-covariance Gaussian VI, low-rank approximations ($\Sigma = aa^\top + D$), or normalizing flows.

PART D: IMPLEMENTATION AND PRACTICAL QUESTIONS

D1. What is the reparameterization trick?

Answer:

In VI, we need to compute gradients of expectations:

$$\nabla_\lambda \mathcal{L}(\lambda) = \nabla_\lambda \mathbb{E}_{z \sim q_\lambda} [f(z)].$$

Problem: If we draw samples $z^{(s)} \sim q_\lambda$, the sampled values do not depend on λ in a differentiable way. Sampling is a non-differentiable operation, so $\partial z^{(s)} / \partial \lambda = 0$ and gradients cannot flow through the sampling step.

Example: Let $q_\lambda(z) = \mathcal{N}(\lambda, 1)$ and $f(z) = z$. Then $\mathbb{E}[z] = \lambda$ and $\frac{d}{d\lambda} \mathbb{E}[z] = 1$. But naively differentiating a sample gives $\frac{\partial}{\partial \lambda} f(z^{(s)}) = f'(z^{(s)}) \cdot \frac{\partial z^{(s)}}{\partial \lambda} = 0$, which is wrong.

Solution (reparameterization trick): Rewrite sampling as a deterministic transformation of parameter-free noise:

$$z = g(\lambda, \epsilon), \quad \epsilon \sim p(\epsilon),$$

where $p(\epsilon)$ does not depend on λ . For the Gaussian case:

$$z = \mu + \sigma \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1).$$

Now z is a differentiable function of (μ, σ) , and gradients flow through the computation graph:

$$\nabla_\lambda \mathcal{L}(\lambda) = \mathbb{E}_\epsilon [\nabla_\lambda f(g(\lambda, \epsilon))].$$

This typically yields much lower-variance gradient estimates than the score-function alternative.

D2. Compare the score-function and pathwise estimators.

Answer:

Aspect	Score-function (REINFORCE)	Pathwise (Reparameterization)
Formula	$\mathbb{E}_{q_\lambda}[f(z)\nabla_\lambda \log q_\lambda(z)]$	$\mathbb{E}_\epsilon[\nabla_\lambda f(g(\lambda, \epsilon))]$
Differentiates	The variational density $\nabla_\lambda \log q_\lambda(z)$	The sample path through $z = g(\lambda, \epsilon)$
Latent types	Discrete and continuous	Continuous only (requires differentiable g)
Variance	Often high; needs baselines/control variates	Typically much lower
Requirements	Can evaluate $\log q_\lambda(z)$ and sample from q_λ	Reparameterizable q_λ and differentiable model

When to prefer score-function: When latent variables are discrete (e.g., categorical), since discrete variables cannot be reparameterized.

When to prefer pathwise: When latent variables are continuous and q_λ is reparameterizable (e.g., Gaussians). The lower variance leads to faster and more stable convergence. This is the standard choice in VAEs.

D3. Why must coordinate ascent updates be cyclic?

Answer:

In mean-field CAVI, each factor q_i is updated by:

$$\log q_i^*(z_i) = \mathbb{E}_{q_{-i}}[\log p(x, z)] + \text{const.}$$

The update for q_i depends on the *current expectations* under all other factors q_{-i} . If all factors were updated simultaneously using stale expectations, the updates would not be self-consistent and the algorithm could oscillate or diverge.

Cyclic (sequential) updates ensure that each factor uses the most recent information from all previously updated factors within the current sweep. This is a block coordinate ascent scheme on the ELBO.

Convergence guarantee: Each coordinate update is guaranteed to *not decrease* the ELBO (since it is the exact optimum of $\mathcal{L}$ with respect to q_i holding q_{-i} fixed). Since the ELBO is bounded above by $\log p(x)$, the sequence of ELBO values is non-decreasing and bounded, so it converges. (Note: convergence is to a local optimum, not necessarily the global optimum.)

PART E: COMPUTATION PROBLEMS

E1: KL divergence between two Bernoulli distributions.

Solution.

For $P = \text{Ber}(p)$ and $Q = \text{Ber}(q)$:

$$\text{KL}(P\|Q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

(a) $P = \text{Ber}(0.8)$, $Q = \text{Ber}(0.5)$:

$$\begin{aligned}\text{KL}(P\|Q) &= 0.8 \ln \frac{0.8}{0.5} + 0.2 \ln \frac{0.2}{0.5} \\ &= 0.8 \ln 1.6 + 0.2 \ln 0.4 \\ &= 0.8(0.4700) + 0.2(-0.9163) \\ &= 0.3760 - 0.1833 = \boxed{0.1927 \text{ nats}}.\end{aligned}$$

(b) $\text{KL}(Q\|P)$ with $Q = \text{Ber}(0.5)$, $P = \text{Ber}(0.8)$:

$$\begin{aligned}\text{KL}(Q\|P) &= 0.5 \ln \frac{0.5}{0.8} + 0.5 \ln \frac{0.5}{0.2} \\ &= 0.5 \ln 0.625 + 0.5 \ln 2.5 \\ &= 0.5(-0.4700) + 0.5(0.9163) \\ &= -0.2350 + 0.4581 = \boxed{0.2231 \text{ nats}}.\end{aligned}$$

(c) $\text{KL}(P\|Q) = 0.1927 \neq 0.2231 = \text{KL}(Q\|P)$, confirming asymmetry.

E2: Entropy and ELBO for a 1D Gaussian model.

Solution.

(a) The joint model is:

$$\log p(x, z) = \log p(x | z) + \log p(z) = -\frac{1}{2}(x - z)^2 - \frac{1}{2}z^2 + C_1.$$

The variational entropy is:

$$H(q) = -\mathbb{E}_q[\log q(z)] = \frac{1}{2} \log(2\pi e \sigma^2).$$

The expected log-joint under $q(z) = \mathcal{N}(\mu, \sigma^2)$:

$$\begin{aligned}\mathbb{E}_q[\log p(x, z)] &= -\frac{1}{2} \mathbb{E}_q[(x - z)^2] - \frac{1}{2} \mathbb{E}_q[z^2] + C_1 \\ &= -\frac{1}{2} [(x - \mu)^2 + \sigma^2] - \frac{1}{2} [\mu^2 + \sigma^2] + C_1.\end{aligned}$$

Therefore:

$$\boxed{\mathcal{L}(\mu, \sigma) = -\frac{1}{2}(x - \mu)^2 - \frac{1}{2}\mu^2 - \sigma^2 + \frac{1}{2} \log(2\pi e \sigma^2) + C_1.}$$

(b) Optimize with respect to μ :

$$\frac{\partial \mathcal{L}}{\partial \mu} = (x - \mu) - \mu = x - 2\mu = 0 \implies \mu^* = \frac{x}{2} = \frac{3}{2}.$$

Optimize with respect to σ^2 (let $v = \sigma^2$):

$$\frac{\partial \mathcal{L}}{\partial v} = -1 + \frac{1}{2v} = 0 \implies v^* = \sigma^{*2} = \frac{1}{2}.$$

So $q^*(z) = \mathcal{N}(3/2, 1/2)$.

(c) The true posterior: since both prior and likelihood are Gaussian, the posterior is conjugate:

$$p(z \mid x) = \mathcal{N}\left(\frac{x}{2}, \frac{1}{2}\right) = \mathcal{N}\left(\frac{3}{2}, \frac{1}{2}\right).$$

This **exactly matches** the variational optimum. In this conjugate Gaussian case, the Gaussian variational family contains the true posterior, so VI recovers it exactly and $\text{KL}(q^* \| p(\cdot \mid x)) = 0$.

E3: Entropy computations.

Solution.

(a) For $P = (0.5, 0.25, 0.125, 0.125)$:

$$\begin{aligned} H(X) &= -[0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.125 \log_2 0.125 + 0.125 \log_2 0.125] \\ &= -[0.5(-1) + 0.25(-2) + 0.125(-3) + 0.125(-3)] \\ &= -[-0.5 - 0.5 - 0.375 - 0.375] \\ &= \boxed{1.75 \text{ bits}}. \end{aligned}$$

(b) For $X \sim \mathcal{N}(\mu, \sigma^2)$ with density $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$:

$$\begin{aligned} h(X) &= -\int_{-\infty}^{\infty} p(x) \log p(x) dx \\ &= -\int p(x) \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2} \right] dx \\ &= \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \underbrace{\mathbb{E}[(X-\mu)^2]}_{=\sigma^2} \\ &= \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2} \\ &= \boxed{\frac{1}{2} \log(2\pi e\sigma^2)}. \end{aligned}$$

This confirms that larger σ gives higher entropy (more spread-out, more uncertain).